

Ach_Speed

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```
#library
library(tidyverse)
library(lme4)
library(lmerTest)
library(MuMIn)
library(performance)
library(readr)
library(dplyr)
library(ggeffects)
library(partR2)
library(slider)
library(tibble)
library(emmeans)
library(scales)

# Vector of animal IDs (as character)
animal_ids <- c("GACH16", "GACH19", "GACH33", "GACH39", "GACH40")

# Base data directory
data_dir <- "../data-raw"

# Empty list to collect all data.frames
data_list <- list()

for (animal in animal_ids) {
  # build path to this animal's top-level folder
  animal_path <- file.path(data_dir, animal)

  # find all "*clean_data.csv" files anywhere under that folder
  csv_files <- list.files(
    path = animal_path,
    pattern = "clean_data\\*.csv$",
    recursive = TRUE,
    full.names = TRUE
  )

  # read each file and store in data_list
  for (file in csv_files) {
    df_name <- paste(animal, gsub(".*/", "", file), sep = "_")
    df_name # inspect the name of the data frame
    suppressMessages(df <- read_csv(file, show_col_types = FALSE))
  }
}
```

```

    # remove any columns that has no header
    df <- df %>% select(-starts_with("..."))
    data_list[[df_name]] <- df
  }
}

# Combine all data frames into one
all_data <- bind_rows(data_list)
# reformat the data
all_data <- all_data %>%
  mutate(lightness = str_extract(session, "(?<=_)[^_]+$"))
all_data <- all_data %>%
  mutate(session = paste(animal, lightness, date, sep = "_"))
all_data$session <- factor(all_data$session)
# remove rows containing NA in velocity or dFF_z
all_data <- all_data %>% drop_na(velocity, dFF_z)
all_data$lightness <- factor(all_data$lightness)
all_data$animal <- factor(all_data$animal)
rm(data_list, df)
all_data$lightness <- relevel(all_data$lightness, ref = "light")
all_data <- all_data %>% rename(speed = velocity) # it is actually velocity, wrong name in the original data
# create session_short variable for easier plotting
session_map <- all_data %>%
  distinct(animal, date, lightness, session) %>% # unique session rows
  arrange(animal, date) %>%
  group_by(animal) %>%
  mutate(day_idx = dense_rank(date)) %>% # same index for same date
  ungroup() %>%
  mutate(
    LD = if_else(lightness == "light", "L", "D"),
    session_facet = paste0(animal, "_", LD, "_", day_idx)
  )

all_data <- all_data %>%
  left_join(session_map %>% select(session, session_facet), by = "session")

all_data$session_facet <- factor(all_data$session_facet)

bin_width <- 0.2 # seconds; should match the GRAB ACh feature

binned_data <- all_data %>%
  arrange(animal, session, time) %>%
  mutate(
    bin = cut(
      time,
      breaks = seq(0, max(time, na.rm = TRUE) + bin_width, by = bin_width),
      right = FALSE,
      labels = FALSE
    ) - 1L
  ) %>%
  group_by(animal, session, lightness, bin) %>%
  summarise(
    # REMOVED time_bin calculation here

```

```

dFF_z      = mean(dFF_z, na.rm = TRUE),
speed = mean(speed, na.rm = TRUE),
x          = mean(x, na.rm = TRUE),
y          = mean(y, na.rm = TRUE),
n_in_bin   = n(),
session_facet = first(session_facet),
.groups = "drop"
) %>%
# ADD time_bin calculation here (it runs on the summarized rows)
mutate(
  time_bin = (bin + 0.5) * bin_width
) %>%
filter(n_in_bin >= 1)
# calculate change of speed (acceleration)
binned_data <- binned_data %>%
  group_by(session) %>%
  arrange(time_bin, .by_group = TRUE) %>%
  mutate(
    # central difference (better than forward difference)
    changeOfSpeed = (lead(speed) - lag(speed)) / (lead(time_bin) - lag(time_bin))
  ) %>%
  ungroup()
binned_data$log_speed <- log(binned_data$speed) # add log speed column
sum(!is.finite(binned_data$log_speed)) # check for log(0)=-Inf

```

```
## [1] 0
```

```

#make session_facet ordered by LD, animal_id, idx
binned_data <- binned_data %>%
  mutate(
    LD = str_match(session_facet, "_(LD)")[,2], # "L" or "D"
    animal_id = str_extract(session_facet, "[^_]+"), # "GACH16"
    idx = as.integer(str_match(session_facet, "_(LD)_([0-9]+)")[,2]) # 1,2,3,...
  ) %>%
  arrange(LD, animal_id, idx) %>%
  mutate(session_facet = factor(session_facet, levels = unique(session_facet)))

```

```

v_threshold <- 3 # slower than 3 cm/s
min_duration_s = 5

# binned_data has: session, time, speed (plus x, y, etc.)
bouts <- binned_data %>%
  arrange(session, time_bin) %>%
  group_by(session) %>%
  mutate(
    slow = speed < 3,
    bout_id = cumsum(slow != lag(slow, default = first(slow)))
  ) %>%
  group_by(session, bout_id) %>%
  summarise(
    slow = first(slow),
    t_start = first(time_bin),
    t_end = last(time_bin),

```

```

    duration_s = as.numeric(t_end - t_start),
    .groups = "drop"
  ) %>%
  filter(slow, duration_s >= min_duration_s)

# Get all original rows that fall inside qualifying bouts
df_slow <- binned_data %>%
  arrange(session, time_bin) %>%
  group_by(session) %>%
  # Re-calculate bout_id on the full dataset
  mutate(
    slow = speed < 3,
    bout_id = cumsum(slow != lag(slow, default = first(slow)))
  ) %>%
  ungroup() %>%
  # ONLY keep rows where the (session, bout_id) pair exists in our filtered 'bouts' list
  semi_join(bouts, by = c("session", "bout_id"))

```

```

max_distance_diameter <- 5 # cm

sampling_rate <- 1 / bin_width
window_size <- round(min_duration_s * sampling_rate)

df_immobile <- df_slow %>%
  arrange(session, time_bin) %>%
  group_by(session) %>%
  mutate(
    # 1) trailing window
    is_slow_window = slider::slide_lgl(
      speed, ~ all(.x < 3, na.rm = TRUE),
      .before = window_size - 1, .complete = TRUE
    ),

    # 2) trailing window <= 5 cm
    is_small_area = slider::slide2_lgl(
      x, y,
      ~ {
        coords <- cbind(.x, .y)
        coords <- coords[stats::complete.cases(coords), , drop = FALSE]
        if (nrow(coords) < 2) return(FALSE) # "immobile"
        max(stats::dist(coords)) <= max_distance_diameter
      },
      .before = window_size - 1, .complete = TRUE
    ),

    # " + "
    valid_end = is_slow_window & is_small_area
  ) %>%
  # " " TRUE window_size
  mutate(
    keep_row = slider::slide_lgl(
      valid_end, ~ any(.x),
      .before = window_size - 1, .complete = FALSE
    )
  )

```

```

    )
  ) %>%
  filter(keep_row) %>%
  select(-is_slow_window, -is_small_area, -valid_end, -keep_row)

# 1. Create a variable for the folder path
output_folder <- "C:/Users/FangJ/OneDrive/ research/Research_Michael_Hasselmo/Ach/Paper/Figures"

# 2. Create the folder if it doesn't exist
if (!dir.exists(output_folder)) {
  dir.create(output_folder)
}

```

1. Visualization

```

# Scatter plot of Velocity vs. dFF_z
set.seed(123) # for reproducibility
p1 <- ggplot(
  binned_data %>% dplyr::sample_frac(0.05),
  aes(x = speed, y = dFF_z, color = lightness)
) +
  geom_point(size = 0.4, alpha = 0.6) +
  labs(
    x = "Speed (cm/s)",
    y = expression(Fluorescence ~ (z-score))
  ) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "speed_vs_dFF.tiff"), # Use .tiff or .pdf
  plot = p1,
  device = "tiff", # Change to "pdf" if needed
  width = 180, # Common width for 2-column journal images
  height = 120, # Adjust height accordingly
  units = "mm", # Use "mm" or "in"
  dpi = 300, # Minimum requirement for most journals
  compression = "lzw", # Reduces TIFF file size without losing quality
  bg = "white" # Set background to white for better visibility
)

# Scatter plot of Speed vs. dFF_z faceted by sessions
p_speed_facet <- ggplot(binned_data, aes(x = speed, y = dFF_z, color = lightness)) +
  geom_point(size=0.5, alpha = 0.6) +
  facet_wrap(~ session_facet) +
  labs(x = "Speed (cm/s)", y = expression(Fluorescence ~ (z-score))) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "speed_vs_dFF_faceted.tiff"),
  plot = p_speed_facet,
  device = "tiff",
  width = 250,
  height = 200,
  units = "mm",

```

```

    dpi = 300,
    compression = "lzw",
    bg = "white"
)
# Scatter plot of Speed vs. dFF_z faceted by animals
p_speed_facet_animal <- ggplot(binned_data, aes(x = speed, y = dFF_z, color = lightness)) +
  geom_point(size=0.2, alpha = 0.2) +
  facet_wrap(~ animal) +
  labs(x = "Speed (cm/s)", y = expression(Fluorescence ~ (z-score))) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "speed_vs_dFF_faceted_by_animal.tiff"),
  plot = p_speed_facet_animal,
  device = "tiff",
  width = 250,
  height = 200,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```

# Scatter plot of Log(speed) vs. dFF_z
p_logspeed <- ggplot(
  binned_data %>% dplyr::sample_frac(0.05),
  aes(x = log_speed, y = dFF_z, color = lightness)
) +
  geom_point(size = 0.4, alpha = 0.6) +
  labs(x = "Log(speed) (cm/s)",
    y = expression(Fluorescence ~ (z-score)))
) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "log_speed_vs_dFF.tiff"),
  plot = p_logspeed,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```

# Scatter plot of Log(speed) vs. dFF_z faceted by sessions
p_logspeed_facet <- ggplot(binned_data, aes(x = log_speed, y = dFF_z, color = lightness)) +
  geom_point(size=0.5, alpha = 0.6) +
  facet_wrap(~ session_facet) +
  labs(x = "Log(speed) (cm/s)", y = expression(Fluorescence ~ (z-score))) +
  theme_minimal()

ggsave(
  filename = file.path(output_folder, "log_speed_vs_dFF_faceted.tiff"),

```

```

plot = p_logspeed_facet,
device = "tiff",
width = 250,
height = 200,
units = "mm",
dpi = 300,
compression = "lzw",
bg = "white"
)

# Scatter plot of Log(speed) vs. dFF_z faceted by animals
p_logspeed_facet_animal <- ggplot(binned_data, aes(x = log_speed, y = dFF_z, color = lightness)) +
  geom_point(size=0.2, alpha = 0.2) +
  facet_wrap(~ animal) +
  labs(x = "Log(speed) (cm/s)", y = expression(Fluorescence ~ (z-score))) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "log_speed_vs_dFF_faceted_by_animal.tiff"),
  plot = p_logspeed_facet_animal,
  device = "tiff",
  width = 250,
  height = 200,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

2. Descriptive statistics

```

# number of observations
n_obs <- nrow(binned_data)
cat("Number of observations:", n_obs, "\n")

```

```
## Number of observations: 166577
```

```

n_imm_obs <- nrow(df_immobile)
cat("Number of immobility observations:", n_imm_obs, "\n")

```

```
## Number of immobility observations: 11077
```

```
cat("Percentage of immobility observations:", round((n_imm_obs / n_obs) * 100, 2), "%\n")
```

```
## Percentage of immobility observations: 6.65 %
```

```

# A) lightness level
tbl_overall_by_lightness <- binned_data %>%
  group_by(lightness) %>%
  summarise(

```

```

    n_animals = n_distinct(animal),
    n_sessions = n_distinct(session),
    n_observations = n(),
    median_obs_per_session = median(as.numeric(table(session))),
    IQR_obs_per_session = IQR(as.numeric(table(session))),
    .groups = "drop"
  )

# B) animal level
tbl_sessions_per_animal <- binned_data %>%
  distinct(animal, session, lightness) %>%
  count(animal, lightness, name = "n_sessions") %>%
  pivot_wider(names_from = lightness, values_from = n_sessions, values_fill = 0) %>%
  mutate(
    has_both_conditions = if_else(rowSums(across(where(is.numeric))) > 0) >= 2, TRUE, FALSE)
  )

# C) session level
tbl_session_level <- binned_data %>%
  group_by(animal, session, lightness) %>%
  summarise(
    n_observations = n(),
    duration_s = max(time_bin, na.rm = TRUE) - min(time_bin, na.rm = TRUE),
    mean_speed = mean(speed, na.rm = TRUE),
    .groups = "drop"
  )

tbl_overall_by_lightness

```

```

## # A tibble: 2 x 6
##   lightness n_animals n_sessions n_observations median_obs_per_session
##   <fct>      <int>      <int>      <int>          <dbl>
## 1 light         5        24        84226          1528
## 2 dark          5        24        82351          1528.
## # i 1 more variable: IQR_obs_per_session <dbl>

```

```
tbl_sessions_per_animal
```

```

## # A tibble: 5 x 4
##   animal light dark has_both_conditions
##   <fct> <int> <int> <lgl>
## 1 GACH16     3     3 TRUE
## 2 GACH19     3     3 TRUE
## 3 GACH33     9     9 TRUE
## 4 GACH39     4     4 TRUE
## 5 GACH40     5     5 TRUE

```

```
summary(tbl_session_level$n_observations)
```

```

##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3056   3097   3190   3470   3529   6430

```



```
summary(tbl_session_level$duration_s)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    611.2   619.3   638.1   694.2   705.9  1286.4
```

```
#comparison at session level
```

```
session_stats <- tbl_session_level %>%
```

```
  group_by(lightness) %>%
```

```
  summarise(
```

```
    mean_duration = mean(duration_s, na.rm = TRUE),
```

```
    sd_duration = sd(duration_s, na.rm = TRUE),
```

```
    mean_speed = mean(mean_speed, na.rm = TRUE),
```

```
    sd_speed = sd(mean_speed, na.rm = TRUE)
```

```
)
```

```
# t-test for speed at session level between light and dark
```

```
t_test_session_speed <- t.test(mean_speed ~ lightness, data = tbl_session_level)
```

```
t_test_session_speed
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: mean_speed by lightness
```

```
## t = -2.4782, df = 45.928, p-value = 0.01694
```

```
## alternative hypothesis: true difference in means between group light and group dark is not equal to 0
```

```
## 95 percent confidence interval:
```

```
## -1.9095190 -0.1977916
```

```
## sample estimates:
```

```
## mean in group light mean in group dark
```

```
##           7.244164           8.297819
```

```
# Heatmaps of occupancy by sessions
```

```
p_occ_heatmap <- ggplot(binned_data, aes(x = x, y = y)) +
```

```
  geom_bin2d(aes(fill = after_stat(count / sum(count))), bins = 60) +
```

```
  scale_fill_viridis_c(name = "Occupancy (fraction)") +
```

```
  facet_wrap(~ session_facet) +
```

```
  coord_equal() +
```

```
  theme_minimal() +
```

```
  labs(x = "X position", y = "Y position")
```

```
ggsave(
```

```
  filename = file.path(output_folder, "occupancy_heatmap_by_session.tiff"),
```

```
  plot = p_occ_heatmap,
```

```
  device = "tiff",
```

```
  width = 250,
```

```
  height = 200,
```

```
  units = "mm",
```

```
  dpi = 300,
```

```
  compression = "lzw",
```

```
  bg = "white"
```

```
)
```

```
# Heatmaps of speed by sessions
```

```

p_speed_heatmap <- ggplot(binned_data, aes(x = x, y = y, z = speed)) +
  stat_summary_2d(
    fun = mean,
    bins = 60
  ) +
  scale_fill_viridis_c(name = "Mean speed") +
  facet_wrap(~ session_facet) +
  coord_equal() +
  theme_minimal() +
  labs(
    x = "X position",
    y = "Y position"
  )
ggsave(
  filename = file.path(output_folder, "speed_heatmap_by_session.tiff"),
  plot = p_speed_heatmap,
  device = "tiff",
  width = 250,
  height = 200,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)
# Heatmaps of dFF_z by sessions
p_dFF_z_heatmap <- ggplot(binned_data, aes(x = x, y = y, z = dFF_z)) +
  stat_summary_2d(
    fun = mean,
    bins = 60
  ) +
  scale_fill_viridis_c(name = expression("Mean " * Fluorescence )) +
  facet_wrap(~ session_facet) +
  coord_equal() +
  theme_minimal() +
  labs(
    x = "X position",
    y = "Y position"
  )
ggsave(
  filename = file.path(output_folder, "dFF_z_heatmap_by_session.tiff"),
  plot = p_dFF_z_heatmap,
  device = "tiff",
  width = 250,
  height = 200,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)
# Histogram of speed
p_hist_speed <- ggplot(binned_data, aes(x = speed, y = after_stat(count/sum(count)), fill = lightness)) +
  geom_histogram(alpha = 0.6, bins = 30, position = "identity") +
  facet_wrap(~ lightness) +

```

```

labs(
  x = "Speed (cm/s)",
  y = "Percentage") +
scale_y_continuous(labels = percent_format(accuracy = 1)) +
theme_minimal()
ggsave(
  filename = file.path(output_folder, "speed_distribution_by_lightness.tiff"),
  plot = p_hist_speed,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)
# Histogram of dFF_z
p_hist_dFF_z <- ggplot(binned_data, aes(x = dFF_z, y = after_stat(count/sum(count)), fill = lightness)) +
  geom_histogram(alpha = 0.6, bins = 30, position = "identity") +
  facet_wrap(~ lightness) +
  labs(
    x = expression(Fluorescence ~ (z-score)),
    y = "Percentage") +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "dFF_z_distribution_by_lightness.tiff"),
  plot = p_hist_dFF_z,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```
summary(binned_data$speed)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## 0.02619  2.33299   5.41740   7.62867 11.16737 42.35675
```

```
summary(binned_data$dFF_z)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## -6.105852 -0.539427  0.113776 -0.000154  0.644221  7.820238
```

```

t_test_speed <- t.test(speed ~ lightness, data = binned_data)
t_test_dFF_z <- t.test(dFF_z ~ lightness, data = binned_data)
cat("T-test for Speed:\n")

```

```
## T-test for Speed:
```

```
print(t_test_speed)
```

```
##  
## Welch Two Sample t-test  
##  
## data: speed by lightness  
## t = -34.77, df = 166507, p-value < 2.2e-16  
## alternative hypothesis: true difference in means between group light and group dark is not equal to 0  
## 95 percent confidence interval:  
## -1.230581 -1.099250  
## sample estimates:  
## mean in group light mean in group dark  
## 7.052767 8.217682
```

```
cat("T-test for dFF_z:\n")
```

```
## T-test for dFF_z:
```

```
print(t_test_dFF_z)
```

```
##  
## Welch Two Sample t-test  
##  
## data: dFF_z by lightness  
## t = -0.002566, df = 166566, p-value = 0.998  
## alternative hypothesis: true difference in means between group light and group dark is not equal to 0  
## 95 percent confidence interval:  
## -0.009250360 0.009226171  
## sample estimates:  
## mean in group light mean in group dark  
## -0.0001601559 -0.0001480613
```

```
# Apply log transformation to speed  
summary(binned_data$log_speed)
```

```
## Min. 1st Qu. Median Mean 3rd Qu. Max.  
## -3.6422 0.8472 1.6896 1.5466 2.4130 3.7461
```

```
t_test_logspeed <- t.test(log_speed ~ lightness, data = binned_data)  
cat("T-test for Log(speed):\n")
```

```
## T-test for Log(speed):
```

```
print(t_test_logspeed)
```

```
##  
## Welch Two Sample t-test  
##  
## data: log_speed by lightness
```

```
## t = -47.342, df = 165733, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group light and group dark is not equal to 0
## 95 percent confidence interval:
## -0.2661097 -0.2449516
## sample estimates:
## mean in group light mean in group dark
## 1.420268 1.675798
```

```
# summary statistics by lightness
```

```
binmed_data %>%
  group_by(lightness) %>%
  summarise(
    mean_dFF_z = mean(dFF_z),
    sd_dFF_z = sd(dFF_z),
    mean_speed = mean(speed),
    sd_speed = sd(speed),
    n = n()
  )
```

```
## # A tibble: 2 x 6
##   lightness mean_dFF_z sd_dFF_z mean_speed sd_speed n
##   <fct>      <dbl>    <dbl>    <dbl>    <dbl> <int>
## 1 light      -0.000160    0.969      7.05     6.84 84226
## 2 dark      -0.000148    0.955      8.22     6.83 82351
```

```
# one-way ANOVA for velocity
```

```
res.aov_velocity <- aov(speed ~ animal, data = binmed_data)
# overall F-test
summary(res.aov_velocity)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## animal         4  297338    74335    1641 <2e-16 ***
## Residuals    166572  7544984      45
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# one-way ANOVA for dFF_z
```

```
res.aov_dFF_z <- aov(dFF_z ~ animal, data = binmed_data)
summary(res.aov_dFF_z)
```

```
##               Df Sum Sq Mean Sq F value Pr(>F)
## animal         4      0    0.0036    0.004      1
## Residuals    166572 154147    0.9254
```

```
# Statistics by animal
```

```
animal_stats <- binmed_data %>%
  group_by(animal) %>%
  summarise(
    count = n(),
    mean_speed = mean(speed, na.rm = TRUE),
    sd_speed = sd(speed, na.rm = TRUE),
    mean_dFF_z = mean(dFF_z, na.rm = TRUE),
```

```
sd_dFF_z = sd(dFF_z, na.rm = TRUE)
)
print(animal_stats)
```

```
## # A tibble: 5 x 6
##   animal count mean_speed sd_speed mean_dFF_z sd_dFF_z
##   <fct> <int>     <dbl>    <dbl>     <dbl>    <dbl>
## 1 GACH16 28659      4.82      4.76 -0.000728  0.977
## 2 GACH19 21627      7.97      6.31 -0.000266  0.985
## 3 GACH33 60111      8.68      7.23  0.00000848 0.972
## 4 GACH39 25301      7.89      6.91 -0.000178  0.914
## 5 GACH40 30879      7.72      7.39  0.000160  0.950
```

3. Regression based analyses

```
# Linear mixed-effects model-Linear Relationship
model_lin <- lmer(dFF_z ~ speed + lightness + (1 | animal/session),
  data = binned_data, REML = FALSE)

# Summary of the model
summary(model_lin)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: dFF_z ~ speed + lightness + (1 | animal/session)
## Data: binned_data
##
##           AIC          BIC      logLik deviance df.resid
## 430210.2 430270.3 -215099.1 430198.2 166571
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.5377 -0.5922  0.0687  0.6607  9.1444
##
## Random effects:
## Groups           Name              Variance Std.Dev.
## session:animal (Intercept) 0.002435 0.04934
## animal          (Intercept) 0.005440 0.07376
## Residual                0.774067 0.87981
## Number of obs: 166577, groups: session:animal, 48; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -4.026e-01 3.483e-02 5.436e+00 -11.560 4.94e-05 ***
## speed        5.836e-02 3.234e-04 1.658e+05 180.451 < 2e-16 ***
## lightnessdark -6.147e-02 1.490e-02 4.261e+01 -4.124 0.000169 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) speed
```

```
## speed          -0.064
## lightnssdrk    -0.212 -0.023
```

```
# 95% confidence intervals for fixed effects
confint(model_lin, parm = "beta_", method = "Wald")
```

```
##                2.5 %      97.5 %
## (Intercept)    -0.47089978 -0.33436572
## speed          0.05772674  0.05899451
## lightnessdark  -0.09067962 -0.03225892
```

```
pred_lin <- ggpredict(model_lin, terms = c("speed [all]", "lightness"))
set.seed(123)
p_linear_sample <- ggplot(pred_lin, aes(x = x, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  labs(
    x = "Speed (cm/s)",
    y = expression(Fluorescence ~ (z-score)),
    color = "Lightness",
    fill = "Lightness"
  ) +
  geom_point(
    data = binned_data %>% dplyr::sample_frac(0.05),
    aes(x = speed, y = dFF_z, color = lightness),
    size = 0.2,
    alpha = 0.2,
    inherit.aes = FALSE
  ) +
  theme_minimal()

ggsave(
  filename = file.path(output_folder, "linear_model_dFF_vs_speed.tiff"),
  plot = p_linear_sample,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)
```

Log transformation of speed

```
# Linear mixed-effects model with log-transformed speed
model_log <- lmer(dFF_z ~ log_speed + lightness + (1 | animal/session),
  data = binned_data, REML = FALSE)

# Summarize the model
summary(model_log)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: dFF_z ~ log_speed + lightness + (1 | animal/session)
## Data: binned_data
##
##      AIC      BIC    logLik deviance df.resid
## 404778.6 404838.7 -202383.3 404766.6   166571
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.8614 -0.6147  0.0317  0.6448  9.8533
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## session:animal (Intercept) 0.008651 0.09301
## animal         (Intercept) 0.014732 0.12138
## Residual              0.664213 0.81499
## Number of obs: 166577, groups: session:animal, 48; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -6.725e-01  5.790e-02  5.477e+00 -11.614 4.59e-05 ***
## log_speed      4.835e-01  1.889e-03  1.665e+05 255.926 < 2e-16 ***
## lightnessdark -1.150e-01  2.716e-02  4.277e+01 -4.235 0.000119 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) lg_spd
## log_speed    -0.045
## lightnssdrk  -0.234 -0.017
```

```
# 95% confidence intervals for fixed effects
confint(model_log, parm = "beta_", method = "Wald")
```

```
##              2.5 %      97.5 %
## (Intercept) -0.7859612 -0.55898332
## log_speed    0.4798438  0.48725012
## lightnessdark -0.1682497 -0.06179298
```

```
# LRT
anova(model_lin, model_log)
```

```
## Data: binned_data
## Models:
## model_lin: dFF_z ~ speed + lightness + (1 | animal/session)
## model_log: dFF_z ~ log_speed + lightness + (1 | animal/session)
##              npar      AIC      BIC logLik deviance Chisq Df Pr(>Chisq)
## model_lin      6 430210 430270 -215099   430198
## model_log      6 404779 404839 -202383   404767 25432  0
```



```
lme4::isSingular(model_log)
```

```
## [1] FALSE
```

```
pred_log <- ggpredict(model_log, terms = c("log_speed [all]", "lightness"))
set.seed(123)
p_log_sample <- ggplot(pred_log, aes(x = x, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  labs(
    x = "Log(speed) (cm/s)",
    y = expression(Fluorescence ~ (z-score)),
    color = "Lightness",
    fill = "Lightness"
  ) +
  geom_point(
    data = binned_data %>% dplyr::sample_frac(0.05),
    aes(x = log_speed, y = dFF_z, color = lightness),
    size = 0.2,
    alpha = 0.2,
    inherit.aes = FALSE
  ) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "log_model_dFF_vs_log_speed.tiff"),
  plot = p_log_sample,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)
```

4. Interaction model

```
model_log_interaction_mid <- lmer(dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed || animal),
  data = binned_data, REML = FALSE)
isSingular(model_log_interaction_mid) # if any random effect is singular
```

```
## [1] FALSE
```

```
VarCorr(model_log_interaction_mid)
```

```
## Groups      Name      Std.Dev.
## animal.session (Intercept) 0.089732
## animal        (Intercept) 0.160454
## animal.1      log_speed   0.087623
## Residual                      0.810624
```

```
anova(model_log, model_log_interaction_mid)
```

```
## Data: binned_data
## Models:
## model_log: dFF_z ~ log_speed + lightness + (1 | animal/session)
## model_log_interaction_mid: dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed || animal)
##               npar      AIC      BIC logLik deviance Chisq Df
## model_log              6 404779 404839 -202383   404767
## model_log_interaction_mid  8 403022 403102 -201503   403006 1760.3  2
##               Pr(>Chisq)
## model_log
## model_log_interaction_mid < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_log_interaction_mid_REML<- update(model_log_interaction_mid,
                                         REML = TRUE)
summary(model_log_interaction_mid_REML)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed ||
##          animal)
## Data: binned_data
##
## REML criterion at convergence: 403028.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.0664 -0.6158  0.0261  0.6436  9.8477
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## animal.session (Intercept) 0.008248 0.09082
## animal          (Intercept) 0.032445 0.18013
## animal.1        log_speed    0.009612 0.09804
## Residual                0.657115 0.81063
## Number of obs: 166577, groups: animal:session, 48; animal, 5
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -6.474e-01  8.298e-02  4.217e+00  -7.802 0.001169 **
## log_speed       4.701e-01  4.392e-02  4.009e+00  10.703 0.000427 ***
## lightnessdark  -1.195e-01  2.720e-02  4.632e+01  -4.392 6.48e-05 ***
## log_speed:lightnessdark  7.858e-03  3.780e-03  1.652e+05   2.079 0.037614 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) lg_spd lghtns
## log_speed    -0.003
## lightnssdrk  -0.162  0.008
## lg_spd:lght   0.030 -0.038 -0.220
```

```
confint(model_log_interaction_mid_REML, parm = "beta_", method = "Wald")
```

```
##                2.5 %      97.5 %
## (Intercept)      -0.8100402440 -0.48478061
## log_speed         0.3839957310  0.55616277
## lightnessdark     -0.1727613003 -0.06614612
## log_speed:lightnessdark 0.0004501404  0.01526635
```

```
# A) slopes for each lightness level
```

```
slopes <- emtrends(model_log_interaction_mid_REML, ~ lightness, var = "log_speed")
slopes
```

```
## lightness log_speed.trend      SE df asymp.LCL asymp.UCL
## light           0.470 0.0439 Inf      0.384      0.556
## dark            0.478 0.0439 Inf      0.392      0.564
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
confint(slopes)
```

```
## lightness log_speed.trend      SE df asymp.LCL asymp.UCL
## light           0.470 0.0439 Inf      0.384      0.556
## dark            0.478 0.0439 Inf      0.392      0.564
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
# B) pairwise comparisons of slopes
```

```
slope_contrast <- pairs(slopes)
slope_contrast
```

```
## contrast      estimate      SE df z.ratio p.value
## light - dark -0.00786 0.00378 Inf  -2.079  0.0376
##
## Degrees-of-freedom method: asymptotic
```

```
confint(slope_contrast)
```

```
## contrast      estimate      SE df asymp.LCL asymp.UCL
## light - dark -0.00786 0.00378 Inf  -0.0153 -0.00045
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
slopes_tbl <- summary(slopes) %>% as.data.frame()
slope_contrast_tbl <- summary(slope_contrast) %>% as.data.frame()
slopes_tbl
```

```
## lightness log_speed.trend      SE  df asymp.LCL asymp.UCL
## light      0.4700793 0.04392097 Inf 0.3839957 0.5561628
## dark      0.4779375 0.04393911 Inf 0.3918184 0.5640566
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
```

```
slope_contrast_tbl
```

```
## contrast      estimate      SE  df z.ratio p.value
## light - dark -0.007858245 0.003779715 Inf  -2.079  0.0376
##
## Degrees-of-freedom method: asymptotic
```

Good, this is not singular, and the interaction is still significant.

```
pred_log_interaction_final <- ggpredict(model_log_interaction_mid_REML, terms = c("log_speed [all]", "lightness"),
set.seed(123)
p_log_interaction <- ggplot(pred_log_interaction_final, aes(x = x, y = predicted, color = group, fill = lightness)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  labs(
    x = "Log(speed) (cm/s)",
    y = expression(Fluorescence ~ (z-score)),
    color = "Lightness",
    fill = "Lightness"
  ) +
  geom_point(
    data = binned_data %>% dplyr::sample_frac(0.05),
    aes(x = log_speed, y = dFF_z, color = lightness),
    size = 0.2,
    alpha = 0.2,
    inherit.aes = FALSE
  ) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "log_interaction_model_dFF_vs_log_speed.tiff"),
  plot = p_log_interaction,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

pred_raw <- as.data.frame(pred_log_interaction_final) %>%
  mutate(
    x = as.numeric(x),
    speed = exp(x)
  )
```

```

# use the raw speed for x-axis but plot the log-based model predictions
set.seed(123)
p_log_interaction_rawx <- ggplot(pred_raw, aes(x = speed, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  geom_point(
    data = binned_data %>% dplyr::sample_frac(0.05),
    aes(x = speed, y = dFF_z, color = lightness),
    size = 0.2, alpha = 0.2,
    inherit.aes = FALSE
  ) +
  labs(
    x = "Speed (cm/s)",
    y = expression(Fluorescence~(z-score)),
    color = "Lightness",
    fill = "Lightness"
  ) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "log_interaction_model_dFF_vs_speed_rawx.tiff"),
  plot = p_log_interaction_rawx,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```

model_lin_interaction_mid <- lmer(dFF_z ~ speed * lightness + (1 | animal:session) + (speed || animal),
  data = binned_data, REML = FALSE)
isSingular(model_lin_interaction_mid) # if any random effect is singular

```

```
## [1] FALSE
```

```
VarCorr(model_lin_interaction_mid)
```

```
## Groups      Name      Std.Dev.
## animal.session (Intercept) 0.051332
## animal        (Intercept) 0.110027
## animal.1      speed      0.026268
## Residual                      0.869758
```

```
anova(model_lin, model_lin_interaction_mid)
```

```

## Data: binned_data
## Models:
## model_lin: dFF_z ~ speed + lightness + (1 | animal/session)
## model_lin_interaction_mid: dFF_z ~ speed * lightness + (1 | animal:session) + (speed || animal)
##               npar      AIC      BIC logLik deviance Chisq Df

```

```
## model_lin          6 430210 430270 -215099  430198
## model_lin_interaction_mid  8 426429 426509 -213206  426413 3785.2  2
##                      Pr(>Chisq)
## model_lin
## model_lin_interaction_mid < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

model_lin_interaction_mid_REML<- update(model_lin_interaction_mid,
                                         REML = TRUE)
summary(model_lin_interaction_mid_REML)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: dFF_z ~ speed * lightness + (1 | animal:session) + (speed ||
##      animal)
##      Data: binned_data
##
## REML criterion at convergence: 426443.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.7099 -0.5957  0.0568  0.6602  9.1864
##
## Random effects:
##      Groups          Name          Variance Std.Dev.
## animal.session (Intercept) 0.0027044 0.05200
## animal         (Intercept) 0.0152592 0.12353
## animal.1        speed       0.0008627 0.02937
## Residual                0.7564837 0.86976
## Number of obs: 166577, groups:  animal:session, 48; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -4.463e-01  5.653e-02  4.148e+00  -7.896  0.00120 **
## speed         6.823e-02  1.314e-02  4.004e+00   5.191  0.00654 **
## lightnessdark -2.347e-02  1.640e-02  5.071e+01  -1.432  0.15840
## speed:lightnessdark -4.621e-03  6.394e-04  1.600e+05  -7.226  4.98e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) speed  lghtns
## speed         -0.002
## lightnssdrk  -0.143  0.007
## spd:lghtnss  0.040 -0.024 -0.303

confint(model_lin_interaction_mid_REML, parm = "beta_", method = "Wald")

##              2.5 %      97.5 %
## (Intercept)  -0.557124947 -0.33554602
## speed         0.042466174  0.09398966
## lightnessdark -0.055611040  0.00866368
## speed:lightnessdark -0.005873998 -0.00336750
```

```

pred_lin_interaction_final <- ggpredict(model_lin_interaction_mid_REML, terms = c("speed [all]", "lightness [all]"),
set.seed(123)
p_lin_interaction <- ggplot(pred_lin_interaction_final, aes(x = x, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  labs(
    x = "Speed (cm/s)",
    y = expression(Fluorescence ~ (z-score)),
    color = "Lightness",
    fill = "Lightness"
  ) +
  geom_point(
    data = binned_data %>% dplyr::sample_frac(0.05),
    aes(x = speed, y = dFF_z, color = lightness),
    size = 0.2,
    alpha = 0.2,
    inherit.aes = FALSE
  ) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "linear_interaction_model_dFF_vs_speed.tiff"),
  plot = p_lin_interaction,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```

dat_cv <- binned_data %>%
  mutate(
    lightness = droplevels(lightness),
    animal = droplevels(animal),
    session = droplevels(session)
  ) %>%
  filter(
    !is.na(dFF_z),
    !is.na(lightness),
    !is.na(animal),
    !is.na(session)
  )

sessions <- levels(dat_cv$session)

score_fold <- function(model, test_df, y = "dFF_z", pred_fixed_only = TRUE) {
  # pred_fixed_only=TRUE only use fixed effects for prediction
  # pred_fixed_only=FALSE use conditional prediction (including random effects of animal; new levels of
  pred <- predict(
    model,
    newdata = test_df,
    re.form = if (pred_fixed_only) NA else NULL,

```

```

    allow.new.levels = TRUE
  )
  err <- test_df[[y]] - pred
  rmse <- sqrt(mean(err^2, na.rm = TRUE))
  mae <- mean(abs(err), na.rm = TRUE)

  # Gaussian log score (the bigger the better)
  sig <- sigma(model)
  logscore <- mean(dnorm(test_df[[y]], mean = pred, sd = sig, log = TRUE), na.rm = TRUE)

  list(rmse = rmse, mae = mae, logscore = logscore)
}

# 1) LOSO leave-one-session-out cross-validation
results_loso <- lapply(sessions, function(s_out) {
  train_df <- dat_cv %>% filter(session != s_out)
  test_df <- dat_cv %>% filter(session == s_out)

  # because we want to compare linear vs. log models fixed effects, let REML=FALSE
  m_lin <- lmer(
    dFF_z ~ speed * lightness + (1 | animal:session) + (speed || animal),
    data = train_df, REML = FALSE
  )

  m_log <- lmer(
    dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed || animal),
    data = train_df, REML = FALSE
  )

  sc_lin <- score_fold(m_lin, test_df, pred_fixed_only = TRUE)
  sc_log <- score_fold(m_log, test_df, pred_fixed_only = TRUE)

  tibble(
    session = s_out,
    n_test = nrow(test_df),

    rmse_lin = sc_lin$rmse,
    rmse_log = sc_log$rmse,
    d_rmse = sc_log$rmse - sc_lin$rmse, # <0 favors log

    mae_lin = sc_lin$mae,
    mae_log = sc_log$mae,
    d_mae = sc_log$mae - sc_lin$mae, # <0 favors log

    lscore_lin = sc_lin$logscore,
    lscore_log = sc_log$logscore,
    d_lscore = sc_log$logscore - sc_lin$logscore # >0 favors log
  )
}) %>% bind_rows()

# 2) summary of LOSO results
summary_loso <- results_loso %>%
  summarise(

```



```

    n_sessions = n(),
    mean_d_rmse = mean(d_rmse),
    sd_d_rmse   = sd(d_rmse),
    mean_d_mae  = mean(d_mae),
    sd_d_mae    = sd(d_mae),
    mean_d_lsore = mean(d_lsore),
    sd_d_lsore  = sd(d_lsore)
  )

print(summary_loso)

```

```

## # A tibble: 1 x 7
##   n_sessions mean_d_rmse sd_d_rmse mean_d_mae sd_d_mae mean_d_lsore sd_d_lsore
##   <int>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1      48    -0.0568    0.0278   -0.0438    0.0184    0.0643    0.0355

```

```

# 3) t.test is actually wrong, but for reference
# d_rmse < 0 --> log and d_lsore > 0 --> log
t_rmse  <- t.test(results_loso$d_rmse, mu = 0)
t_mae   <- t.test(results_loso$d_mae,  mu = 0)
t_lsore <- t.test(results_loso$d_lsore, mu = 0)

t_rmse

```

```

##
## One Sample t-test
##
## data: results_loso$d_rmse
## t = -14.139, df = 47, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -0.06483831 -0.04868603
## sample estimates:
## mean of x
## -0.05676217

```

```
t_mae
```

```

##
## One Sample t-test
##
## data: results_loso$d_mae
## t = -16.484, df = 47, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -0.04909238 -0.03841284
## sample estimates:
## mean of x
## -0.04375261

```

```
t_lsore
```

```
##  
## One Sample t-test  
##  
## data: results_loso$d_lsore  
## t = 12.545, df = 47, p-value < 2.2e-16  
## alternative hypothesis: true mean is not equal to 0  
## 95 percent confidence interval:  
## 0.05402819 0.07466672  
## sample estimates:  
## mean of x  
## 0.06434745
```

```
# 4) bootstrap 95% CI for mean differences
```

```
set.seed(1)  
B <- 5000  
boot_mean_drmse <- replicate(B, mean(sample(results_loso$d_rmse, replace = TRUE)))  
boot_ci_drmse <- quantile(boot_mean_drmse, c(0.025, 0.975))  
  
boot_mean_dlscore <- replicate(B, mean(sample(results_loso$d_lsore, replace = TRUE)))  
boot_ci_dlscore <- quantile(boot_mean_dlscore, c(0.025, 0.975))  
  
boot_ci_drmse
```

```
##          2.5%          97.5%  
## -0.06473488 -0.04916516
```

```
boot_ci_dlscore
```

```
##          2.5%          97.5%  
## 0.05441121 0.07441461
```

```
# effect size
```

```
r.squaredGLMM(model_lin)
```

```
##          R2m          R2c  
## [1,] 0.1691618 0.1775294
```

```
r.squaredGLMM(model_log)
```

```
##          R2m          R2c  
## [1,] 0.2924875 0.3165479
```

```
r.squaredGLMM(model_lin_interaction_mid_REML)
```

```
##          R2m          R2c  
## [1,] 0.1904953 0.2922652
```

```
r.squaredGLMM(model_log_interaction_mid_REML)
```

```
##           R2m           R2c
## [1,] 0.2712692 0.3463907
```

```
AIC(model_lin, model_log)
```

```
##           df           AIC
## model_lin  6 430210.2
## model_log  6 404778.6
```

```
AIC(model_lin_interaction_mid, model_log_interaction_mid)
```

```
##           df           AIC
## model_lin_interaction_mid  8 426429.0
## model_log_interaction_mid  8 403022.3
```

```
binned_data_clear <- binned_data %>%
  drop_na(changeOfSpeed)
model_log_only <- lmer(dFF_z ~ log_speed + lightness + (1 | animal:session) + (log_speed || animal),
  data = binned_data_clear, REML = FALSE, na.action = na.omit)
model_ChangeOfSpeedAndLog <- lmer(dFF_z ~ log_speed * lightness + changeOfSpeed * lightness + (1 | animal:session) + (log_speed || animal) + (changeOfSpeed || animal),
  data = binned_data_clear, REML = FALSE, na.action = na.omit)
summary(model_ChangeOfSpeedAndLog)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: dFF_z ~ log_speed * lightness + changeOfSpeed * lightness + (1 |
## animal:session) + (log_speed || animal) + (changeOfSpeed || animal)
## Data: binned_data_clear
##
##           AIC           BIC      logLik  deviance  df.resid
## 399209.8 399330.0 -199592.9 399185.8    166469
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.1648 -0.6166  0.0242  0.6425  9.9342
##
## Random effects:
## Groups           Name              Variance Std.Dev.
## animal.session (Intercept) 8.066e-03 0.089809
## animal          (Intercept) 1.121e-01 0.334775
## animal.1        log_speed    4.780e-02 0.218623
## animal.2        (Intercept) 1.209e+00 1.099354
## animal.3        changeOfSpeed 1.565e-06 0.001251
## Residual                    6.429e-01 0.801831
## Number of obs: 166481, groups: animal:session, 48; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -6.472e-01 5.143e-01 1.474e+05 -1.258 0.2082
```

```
## log_speed          4.702e-01  9.781e-02  1.664e+05   4.808 1.53e-06
## lightnessdark     -1.190e-01  2.690e-02  4.750e+01  -4.425 5.61e-05
## changeOfSpeed     -8.748e-03  6.015e-04  5.808e+00 -14.544 8.66e-06
## log_speed:lightnessdark  7.517e-03  3.740e-03  1.651e+05   2.010  0.0445
## lightnessdark:changeOfSpeed 4.461e-04  2.826e-04  1.656e+05   1.578  0.1145
##
## (Intercept)
## log_speed          ***
## lightnessdark      ***
## changeOfSpeed       ***
## log_speed:lightnessdark  *
## lightnessdark:changeOfSpeed
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) lg_spd lghtns chngOS lg_sp:
## log_speed    0.000
## lightnssdrk -0.026  0.003
## changeOfSpd  0.000  0.000  0.000
## lg_spd:lght  0.005 -0.017 -0.221  0.000
## lghtnssd:OS  0.000  0.000  0.000 -0.244  0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate Hessian with 2 negative eigenvalues
```

```
confint(model_ChangeOfSpeedAndLog, parm = "beta_", method = "Wald")
```

```
##                  2.5 %      97.5 %
## (Intercept)      -1.6552560829  0.360817990
## log_speed         0.2785412298  0.661930243
## lightnessdark     -0.1717491600 -0.066313425
## changeOfSpeed     -0.0099274426 -0.007569548
## log_speed:lightnessdark  0.0001855604  0.014848051
## lightnessdark:changeOfSpeed -0.0001078233  0.001000120
```

```
anova(model_log_only, model_ChangeOfSpeedAndLog)
```

```
## Data: binned_data_clear
## Models:
## model_log_only: dFF_z ~ log_speed + lightness + (1 | animal:session) + (log_speed || animal)
## model_ChangeOfSpeedAndLog: dFF_z ~ log_speed * lightness + changeOfSpeed * lightness + (1 | animal:session)
##              npar    AIC    BIC logLik deviance Chisq Df
## model_log_only      7 402738 402809 -201362   402724
## model_ChangeOfSpeedAndLog 12 399210 399330 -199593   399186 3538.7  5
##              Pr(>Chisq)
## model_log_only
## model_ChangeOfSpeedAndLog < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

model_absChangeOfSpeedAndLog <- lmer(dFF_z ~ log_speed * lightness + abs(changeOfSpeed) * lightness + (
  data = binned_data_clear, REML = FALSE, na.action = na.omit)
summary(model_absChangeOfSpeedAndLog)

```

```

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: dFF_z ~ log_speed * lightness + abs(changeOfSpeed) * lightness +
## (1 | animal:session) + (log_speed || animal) + (abs(changeOfSpeed) ||
## animal)
## Data: binned_data_clear
##
##           AIC          BIC      logLik  deviance  df.resid
## 402620.3  402740.6 -201298.2  402596.3    166469
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.0593 -0.6166  0.0266  0.6443  9.8519
##
## Random effects:
## Groups          Name              Variance Std.Dev.
## animal.session (Intercept)      8.043e-03 0.089684
## animal         (Intercept)      2.077e+00 1.441329
## animal.1       log_speed        9.641e-03 0.098189
## animal.2       (Intercept)      3.714e-01 0.609429
## animal.3       abs(changeOfSpeed) 1.385e-05 0.003722
## Residual                          6.562e-01 0.810088
## Number of obs: 166481, groups: animal:session, 48; animal, 5
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)   -6.527e-01  7.001e-01 8.482e+04  -0.932
## log_speed      4.625e-01  4.402e-02 3.813e+00  10.506
## lightnessdark -1.202e-01  2.691e-02 4.841e+01  -4.467
## abs(changeOfSpeed) 2.105e-03  1.704e-03 4.988e+00   1.235
## log_speed:lightnessdark 6.032e-03  4.350e-03 1.660e+05   1.387
## lightnessdark:abs(changeOfSpeed) 4.185e-04  4.669e-04 1.664e+05   0.896
##
##              Pr(>|t|)
## (Intercept)    0.351171
## log_speed      0.000593 ***
## lightnessdark  4.75e-05 ***
## abs(changeOfSpeed) 0.271766
## log_speed:lightnessdark 0.165543
## lightnessdark:abs(changeOfSpeed) 0.370069
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) lg_spd lghtns ab(OS) lg_sp:
## log_speed      0.000
## lightnessdark -0.019  0.007
## abs(chngOS)    0.000 -0.008  0.007
## lg_spd:lght    0.003 -0.046 -0.171  0.069
## lghtns:(OS)    0.001  0.025 -0.045 -0.140 -0.495

```

```
## optimizer (nloptwrap) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate Hessian with 2 negative eigenvalues
```

```
confint(model_absChangeOfSpeedAndLog, parm = "beta_", method = "Wald")
```

```
##                2.5 %      97.5 %
## (Intercept)    -2.0249227824  0.719461290
## log_speed      0.3761854070  0.548728918
## lightnessdark  -0.1729638254 -0.067477112
## abs(changeOfSpeed) -0.0012349404  0.005444052
## log_speed:lightnessdark -0.0024935867  0.014556553
## lightnessdark:abs(changeOfSpeed) -0.0004966024  0.001333611
```

```
anova(model_log_only, model_absChangeOfSpeedAndLog)
```

```
## Data: binned_data_clear
## Models:
## model_log_only: dFF_z ~ log_speed + lightness + (1 | animal:session) + (log_speed || animal)
## model_absChangeOfSpeedAndLog: dFF_z ~ log_speed * lightness + abs(changeOfSpeed) * lightness + (1 | animal)
##               npar    AIC    BIC logLik deviance Chisq Df
## model_log_only      7 402738 402809 -201362   402724
## model_absChangeOfSpeedAndLog 12 402620 402741 -201298   402596 128.13  5
##               Pr(>Chisq)
## model_log_only
## model_absChangeOfSpeedAndLog < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
check_collinearity(model_ChangeOfSpeedAndLog)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##               Term  VIF      VIF 95% CI Increased SE Tolerance
##               log_speed 1.00 [1.00, 4868.20]      1.00      1.00
##               lightness 1.05 [1.05, 1.06]      1.03      0.95
##               changeOfSpeed 1.06 [1.06, 1.07]      1.03      0.94
##               log_speed:lightness 1.05 [1.05, 1.06]      1.03      0.95
##               lightness:changeOfSpeed 1.06 [1.06, 1.07]      1.03      0.94
## Tolerance 95% CI
##               [0.00, 1.00]
##               [0.95, 0.96]
##               [0.94, 0.94]
##               [0.95, 0.96]
##               [0.94, 0.94]
```

```
check_collinearity(model_absChangeOfSpeedAndLog)
```

```
## # Check for Multicollinearity
```

```
##
## Low Correlation
##
##           Term  VIF  VIF 95% CI Increased SE Tolerance
##           log_speed 1.00 [1.00, 1.02]          1.00      1.00
##           lightness 1.05 [1.05, 1.06]          1.03      0.95
##           abs(changeOfSpeed) 1.02 [1.02, 1.03]          1.01      0.98
##           log_speed:lightness 1.40 [1.39, 1.40]          1.18      0.72
##           lightness:abs(changeOfSpeed) 1.38 [1.37, 1.38]          1.17      0.73
## Tolerance 95% CI
##           [0.98, 1.00]
##           [0.94, 0.95]
##           [0.97, 0.98]
##           [0.71, 0.72]
##           [0.72, 0.73]
```

```
# effect size
model_ChangeOfSpeedAndLog_REML <- update(model_ChangeOfSpeedAndLog, REML = TRUE)
r.squaredGLMM(model_ChangeOfSpeedAndLog_REML)
```

```
##           R2m      R2c
## [1,] 0.05339266 0.8802804
```

```
model_absChangeOfSpeedAndLog_REML <- update(model_absChangeOfSpeedAndLog, REML = TRUE)
r.squaredGLMM(model_absChangeOfSpeedAndLog_REML)
```

```
##           R2m      R2c
## [1,] 0.1293806 0.6921993
```

```
model_log_only_REML <- update(model_log_only, REML = TRUE)
r.squaredGLMM(model_log_only_REML)
```

```
##           R2m      R2c
## [1,] 0.2712202 0.3465869
```

```
AIC(model_log_only, model_ChangeOfSpeedAndLog, model_absChangeOfSpeedAndLog)
```

```
##           df      AIC
## model_log_only      7 402738.4
## model_ChangeOfSpeedAndLog 12 399209.8
## model_absChangeOfSpeedAndLog 12 402620.3
```

5. regression with immobility

```
imm_linear_interaction_mid <- lmer(dFF_z ~ speed * lightness + (1 | animal:session) + (speed || animal)
  data = df_immobile, REML = FALSE)
imm_log_interaction_mid <- lmer(dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed || animal)
  data = df_immobile, REML = FALSE)
```

```
summary(imm_linear_interaction_mid)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: dFF_z ~ speed * lightness + (1 | animal:session) + (speed ||
## animal)
## Data: df_immobile
##
##      AIC      BIC    logLik deviance df.resid
## 28668.2 28726.7 -14326.1 28652.2    11069
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.6908 -0.6525 -0.0200  0.6535  4.8887
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## animal.session (Intercept) 0.42140  0.6492
## animal         (Intercept) 0.12723  0.3567
## animal.1       speed        0.04446  0.2109
## Residual                          0.76277  0.8734
## Number of obs: 11077, groups: animal:session, 46; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -2.042e+00  2.129e-01  8.028e+00  -9.589 1.13e-05 ***
## speed          4.239e-01  9.592e-02  4.970e+00   4.419  0.0070 **
## lightnessdark  -3.757e-01  1.997e-01  4.392e+01  -1.881  0.0666 .
## speed:lightnessdark -5.313e-02  2.884e-02  1.107e+04  -1.842  0.0654 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) speed  lgtns
## speed        -0.019
## lightnssdrk  -0.433  0.015
## spd:lgtnss   0.045 -0.084 -0.177
```

```
summary(imm_log_interaction_mid)
```

```
## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed ||
## animal)
## Data: df_immobile
##
##      AIC      BIC    logLik deviance df.resid
## 28436.9 28495.4 -14210.5 28420.9    11069
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.7501 -0.6442 -0.0139  0.6533  4.8997
```



```
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## animal.session (Intercept) 0.42008 0.6481
## animal        (Intercept) 0.06756 0.2599
## animal.1      log_speed   0.02770 0.1664
## Residual                      0.74720 0.8644
## Number of obs: 11077, groups: animal:session, 46; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -1.540e+00  1.807e-01  1.062e+01  -8.522 4.53e-06 ***
## log_speed      3.910e-01  7.633e-02  4.688e+00   5.123  0.00444 **
## lightnessdark  -4.508e-01  1.960e-01  4.208e+01  -2.300  0.02648 *
## log_speed:lightnessdark -3.701e-02  2.500e-02  1.105e+04  -1.480  0.13883
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) lg_spd lghtns
## log_speed    -0.002
## lightnssdrk  -0.509 -0.001
## lg_spd:lght  -0.002 -0.081 -0.003
```

```
anova(imm_linear_interaction_mid, imm_log_interaction_mid)
```

```
## Data: df_immobile
## Models:
## imm_linear_interaction_mid: dFF_z ~ speed * lightness + (1 | animal:session) + (speed || animal)
## imm_log_interaction_mid: dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed || animal)
##              npar    AIC    BIC logLik deviance Chisq Df
## imm_linear_interaction_mid    8 28668 28727 -14326    28652
## imm_log_interaction_mid      8 28437 28495 -14210    28421 231.22  0
##              Pr(>Chisq)
## imm_linear_interaction_mid
## imm_log_interaction_mid
```

```
isSingular(imm_log_interaction_mid)
```

```
## [1] FALSE
```

```
confint(imm_log_interaction_mid, parm = "beta_", method = "Wald")
```

```
##              2.5 %      97.5 %
## (Intercept)   -1.89396201 -1.18566406
## log_speed      0.24140137  0.54060837
## lightnessdark  -0.83497985 -0.06663778
## log_speed:lightnessdark -0.08601429  0.01199389
```

```
confint(imm_linear_interaction_mid, parm = "beta_", method = "Wald")
```

```
##              2.5 %      97.5 %
## (Intercept)   -2.4588974 -1.624307660
## speed         0.2358711  0.611866110
## lightnessdark -0.7671782  0.015743487
## speed:lightnessdark -0.1096577  0.003387744
```

```
imm_log_interaction_mid_REML <- update(imm_log_interaction_mid, REML = TRUE)
summary(imm_log_interaction_mid_REML)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed ||
##      animal)
##      Data: df_immobile
##
## REML criterion at convergence: 28432.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.7482 -0.6436 -0.0140  0.6530  4.8988
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## animal.session (Intercept) 0.43317  0.6582
## animal         (Intercept) 0.09126  0.3021
## animal.1       log_speed    0.03553  0.1885
## Residual                        0.74726  0.8644
## Number of obs: 11077, groups:  animal:session, 46; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -1.533e+00  1.951e-01  7.822e+00  -7.858 5.62e-05 ***
## log_speed      3.899e-01  8.598e-02  3.803e+00   4.535  0.0118 *
## lightnessdark  -4.505e-01  1.990e-01  4.077e+01  -2.264  0.0290 *
## log_speed:lightnessdark -3.723e-02  2.501e-02  1.105e+04  -1.489  0.1366
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) lg_spd lghtns
## log_speed    -0.001
## lightnssdrk  -0.478  0.000
## lg_spd:lght  -0.002 -0.072 -0.003
```

```
imm_linear_interaction_mid_REML <- update(imm_linear_interaction_mid, REML = TRUE)
summary(imm_linear_interaction_mid_REML)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: dFF_z ~ speed * lightness + (1 | animal:session) + (speed ||
##      animal)
##      Data: df_immobile
##
```

```
## REML criterion at convergence: 28662.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.6899 -0.6524 -0.0201  0.6531  4.8876
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## animal.session (Intercept) 0.43337   0.6583
## animal          (Intercept) 0.17035   0.4127
## animal.1        speed         0.05644   0.2376
## Residual                                0.76283   0.8734
## Number of obs: 11077, groups:  animal:session, 46; animal, 5
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -2.038e+00  2.336e-01  5.969e+00  -8.723 0.000129 ***
## speed          4.234e-01  1.077e-01  3.981e+00   3.932 0.017236 *
## lightnessdark  -3.734e-01  2.024e-01  4.261e+01  -1.844 0.072094 .
## speed:lightnessdark -5.340e-02  2.884e-02  1.106e+04  -1.851 0.064135 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) speed  lgtns
## speed          -0.015
## lightnssdrk    -0.400  0.013
## spd:lgtnss     0.041 -0.075 -0.175
```

```
# effect size
r.squaredGLMM(imm_linear_interaction_mid_REML)
```

```
##              R2m       R2c
## [1,] 0.07538156 0.5124225
```

```
r.squaredGLMM(imm_log_interaction_mid_REML)
```

```
##              R2m       R2c
## [1,] 0.09143464 0.4758873
```

```
AIC(imm_linear_interaction_mid, imm_log_interaction_mid)
```

```
##              df      AIC
## imm_linear_interaction_mid  8 28668.17
## imm_log_interaction_mid     8 28436.94
```

```
pred_lin <- ggpredict(imm_linear_interaction_mid_REML, terms = c("speed [all]", "lightness"))
set.seed(123)
p_imm_linear_sample <- ggplot(pred_lin, aes(x = x, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
```

```

labs(
  x = "Speed (cm/s)",
  y = expression(Fluorescence ~ (z-score)),
  color = "Lightness",
  fill = "Lightness"
) +
geom_point(
  data = df_immobile %>% dplyr::sample_frac(0.05),
  aes(x = speed, y = dFF_z, color = lightness),
  size = 0.2,
  alpha = 0.2,
  inherit.aes = FALSE
) +
theme_minimal()

ggsave(
  filename = file.path(output_folder, "imm_linear_interaction_model_dFF_vs_speed.tiff"),
  plot = p_imm_linear_sample,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```

pred_log_interaction_final <- ggpredict(imm_log_interaction_mid_REML, terms = c("log_speed [all]", "lightness [all]"),
set.seed(123)
p_imm_log_interaction <- ggplot(pred_log_interaction_final, aes(x = x, y = predicted, color = group, fill = group)) +
  geom_line(linewidth = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.15, color = NA) +
  labs(
    x = "Log(speed) (cm/s)",
    y = expression(Fluorescence ~ (z-score)),
    color = "Lightness",
    fill = "Lightness"
  ) +
  geom_point(
    data = df_immobile %>% dplyr::sample_frac(0.2),
    aes(x = log_speed, y = dFF_z, color = lightness),
    size = 0.2,
    alpha = 0.2,
    inherit.aes = FALSE
  ) +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "imm_log_interaction_model_dFF_vs_log_speed.tiff"),
  plot = p_imm_log_interaction,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",

```

```

dpi = 300,
compression = "lzw",
bg = "white"
)

dat_cv <- df_immobile %>%
  mutate(
    lightness = droplevels(lightness),
    animal = droplevels(animal),
    session = droplevels(session)
  ) %>%
  filter(
    !is.na(dFF_z),
    !is.na(lightness),
    !is.na(animal),
    !is.na(session)
  )

sessions <- levels(dat_cv$session)

score_fold <- function(model, test_df, y = "dFF_z", pred_fixed_only = TRUE) {
  # pred_fixed_only=TRUE only use fixed effects for prediction
  # pred_fixed_only=FALSE use conditional prediction (including random effects of animal; new levels of
  pred <- predict(
    model,
    newdata = test_df,
    re.form = if (pred_fixed_only) NA else NULL,
    allow.new.levels = TRUE
  )
  err <- test_df[[y]] - pred
  rmse <- sqrt(mean(err^2, na.rm = TRUE))
  mae <- mean(abs(err), na.rm = TRUE)

  # Gaussian log score (the bigger the better)
  sig <- sigma(model)
  logscore <- mean(dnorm(test_df[[y]], mean = pred, sd = sig, log = TRUE), na.rm = TRUE)

  list(rmse = rmse, mae = mae, logscore = logscore)
}

# 1) LOSO leave-one-session-out cross-validation
results_loso <- lapply(sessions, function(s_out) {
  train_df <- dat_cv %>% filter(session != s_out)
  test_df <- dat_cv %>% filter(session == s_out)

  # because we want to compare linear vs. log models fixed effects, let REML=FALSE
  m_lin <- lmer(
    dFF_z ~ speed * lightness + (1 | animal:session) + (speed || animal),
    data = train_df, REML = FALSE
  )

  m_log <- lmer(
    dFF_z ~ log_speed * lightness + (1 | animal:session) + (log_speed || animal),

```

```

    data = train_df, REML = FALSE
  )

  sc_lin <- score_fold(m_lin, test_df, pred_fixed_only = TRUE)
  sc_log <- score_fold(m_log, test_df, pred_fixed_only = TRUE)

  tibble(
    session = s_out,
    n_test  = nrow(test_df),

    rmse_lin = sc_lin$rmse,
    rmse_log = sc_log$rmse,
    d_rmse   = sc_log$rmse - sc_lin$rmse,      # <0 favors log

    mae_lin  = sc_lin$mae,
    mae_log  = sc_log$mae,
    d_mae    = sc_log$mae - sc_lin$mae,      # <0 favors log

    lscore_lin = sc_lin$logscore,
    lscore_log = sc_log$logscore,
    d_lscore   = sc_log$logscore - sc_lin$logscore # >0 favors log
  )
}) %>% bind_rows()

# 2) summary of LOSO results
summary_loso <- results_loso %>%
  summarise(
    n_sessions = n(),
    mean_d_rmse = mean(d_rmse),
    sd_d_rmse   = sd(d_rmse),
    mean_d_mae  = mean(d_mae),
    sd_d_mae    = sd(d_mae),
    mean_d_lscore = mean(d_lscore),
    sd_d_lscore  = sd(d_lscore)
  )

print(summary_loso)

## # A tibble: 1 x 7
##   n_sessions mean_d_rmse sd_d_rmse mean_d_mae sd_d_mae mean_d_lscore sd_d_lscore
##   <int>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1       46    -0.00111  0.0373   0.000564  0.0372   -0.000923  0.0504

# 3) t.test is actually wrong, but for reference
# d_rmse < 0 --> log and d_lscore > 0 --> log
t_rmse <- t.test(results_loso$d_rmse, mu = 0)
t_mae <- t.test(results_loso$d_mae, mu = 0)
t_lscore <- t.test(results_loso$d_lscore, mu = 0)

t_rmse

##
## One Sample t-test

```

```
##
## data: results_loso$d_rmse
## t = -0.20199, df = 45, p-value = 0.8408
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -0.012185488 0.009964169
## sample estimates:
## mean of x
## -0.00111066
```

```
t_mae
```

```
##
## One Sample t-test
##
## data: results_loso$d_mae
## t = 0.1028, df = 45, p-value = 0.9186
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -0.01048275 0.01161040
## sample estimates:
## mean of x
## 0.0005638248
```

```
t_lscore
```

```
##
## One Sample t-test
##
## data: results_loso$d_lscore
## t = -0.12407, df = 45, p-value = 0.9018
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## -0.01589847 0.01405337
## sample estimates:
## mean of x
## -0.0009225505
```

```
# 4) bootstrap 95% CI for mean differences
```

```
set.seed(1)
B <- 5000
boot_mean_drmse <- replicate(B, mean(sample(results_loso$d_rmse, replace = TRUE)))
boot_ci_drmse <- quantile(boot_mean_drmse, c(0.025, 0.975))

boot_mean_dlscore <- replicate(B, mean(sample(results_loso$d_lscore, replace = TRUE)))
boot_ci_dlscore <- quantile(boot_mean_dlscore, c(0.025, 0.975))

boot_ci_drmse
```

```
##          2.5%          97.5%
## -0.01064194 0.01056788
```

```
boot_ci_dlscore
```

```
##          2.5%          97.5%  
## -0.01472800  0.01446187
```

6. Methods for examining regression assumptions

```
binmed_data$fitmed_values <- fitted(model_log_interaction_mid_REML)  
binmed_data$residuals <- residuals(model_log_interaction_mid_REML)  
  
# Scatter plot of residuals vs. fitted values  
p_residuals_vs_fitted <- ggplot(binmed_data, aes(x = fitmed_values, y = residuals, color = lightness)) +  
  geom_point(size=0.2, alpha = 0.2) +  
  labs(x = "Fitted Fluorescence (z-score)", y = "Residuals") +  
  theme_minimal()  
ggsave(  
  filename = file.path(output_folder, "residuals_vs_fitted.tiff"),  
  plot = p_residuals_vs_fitted,  
  device = "tiff",  
  width = 180,  
  height = 150,  
  units = "mm",  
  dpi = 300,  
  compression = "lzw",  
  bg = "white"  
)  
  
# Scatter plot of residuals vs. log_speed  
p_residuals_vs_log_speed <- ggplot(binmed_data, aes(x = log_speed, y = residuals, color = lightness)) +  
  geom_point(size=0.2, alpha = 0.2) +  
  geom_smooth(se = FALSE) +  
  labs(x = "Log(speed) (cm/s)", y = "Residuals") +  
  theme_minimal()  
ggsave(  
  filename = file.path(output_folder, "residuals_vs_log_speed.tiff"),  
  plot = p_residuals_vs_log_speed,  
  device = "tiff",  
  width = 180,  
  height = 150,  
  units = "mm",  
  dpi = 300,  
  compression = "lzw",  
  bg = "white"  
)  
  
# Scatter plot of observed vs. fitted values  
p_observed_vs_fitted <- ggplot(binmed_data, aes(x = fitmed_values, y = dFF_z, color = lightness)) +  
  geom_point(size=0.2, alpha = 0.2) +  
  geom_smooth(method = "lm", se = FALSE, color = "black") +  
  labs(x = "Fitted Fluorescence (z-score)",  
       y = "Observed Fluorescence (z-score)") +  
  theme_minimal()  
ggsave(  
  filename = file.path(output_folder, "observed_vs_fitted.tiff"),  
  plot = p_observed_vs_fitted,  
  device = "tiff",  
  width = 180,  
  height = 150,  
  units = "mm",  
  dpi = 300,  
  compression = "lzw",  
  bg = "white"  
)
```



```

filename = file.path(output_folder, "observed_vs_fitted.tiff"),
plot = p_observed_vs_fitted,
device = "tiff",
width = 180,
height = 150,
units = "mm",
dpi = 300,
compression = "lzw",
bg = "white"
)
# Q-Q plot of residuals
p_qq_plot <- ggplot(binned_data, aes(sample = residuals)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_minimal()
ggsave(
  filename = file.path(output_folder, "qq_plot_residuals.tiff"),
  plot = p_qq_plot,
  device = "tiff",
  width = 180,
  height = 150,
  units = "mm",
  dpi = 300,
  compression = "lzw",
  bg = "white"
)

```

```

# compute lag-1 autocorrelation per session, then summarize
acf_by_session <- binned_data %>%
  group_by(animal, session) %>%
  arrange(time_bin, .by_group = TRUE) %>%
  summarise(
    acf1 = cor(residuals[-1], residuals[-n()], use = "complete.obs"),
    .groups = "drop"
  )

summary(acf_by_session$acf1)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5476  0.8699  0.8834  0.8712  0.9015  0.9393

```

It is highly autocorrelated, which is not surprising but not ideal for LMM. But given the complexity of the data structure, it is hard to model the autocorrelation properly, and other researchers have also used LMM for similar data.